fields $\boldsymbol{f}_i = \boldsymbol{J}_i \nabla H$. Each of these vector fields preserves H and, because they are themselves Hamiltonian, their flows are also volume-preserving. Assuming each such vector field is exactly integrable, the system dynamics may be approximated by composing the flow maps for each vector field.

As an example of such a J-splitting, we may take $K = \mathrm{N_c}$ with each $\boldsymbol{J}_i$ to be defined by the components of the canonical symplectic structure matrix having to do with the ith position and momentum pair, thus the equations of motion for the ith vector field become

$$\dot{q}_i = \frac{p_i}{m_i}, \quad \dot{p}_i = -\frac{\partial U}{\partial q_i}(\bar{q}_1, \ldots, \bar{q}_{i-1}, q_i, \bar{q}_{i+1}, \ldots, \bar{q}_{\mathrm{N_c}}),$$

where components with a bar over them are fixed during the calculation. This is, itself, a Hamiltonian system with a single degree of freedom q_i and the potential $\hat{U}_i(q_i)$ obtained by freezing all other components in the original potential function. Any such 1-dof Hamiltonian system is formally integrable (albeit in quadratures). Thus, if we are willing to solve a system using quadrature, we have a numerical method for the Hamiltonian system which preserves volume and energy. We solve each such dynamics from $i = 1$ to $\mathrm{N_c}$, updating the values of each (q_i, p_i) in sequence (the temporally contiguous nature of the updates rules it out as a strategy for parallel computation).

Although this method can be applied to molecular systems, its implementation is challenging and it is not recommended for general computation. Some systems may have a structure amenable to exploitation via this splitting strategy, or it could potentially serve as a starting point in some special purpose methods. For another use of J-splitting in the setting of Metropolis correction of deterministic methods, see [48].

7.7 The Evolution of the Distribution Using a Numerical Method

In this section, we shall give an approach that allows the computation of the errors introduced by the discretization of Langevin dynamics for more general potential energy functions. The discretization schemes we shall consider will be those created by composition of the A, B and O splitting pieces described earlier in this chapter, but in principle we can apply this analysis to a wider class of SDE methods. We outline the basic approach for determining the leading order behavior of the discretization error introduced by our SDE integration scheme (see [221–223]), based on expansions due to Talay and Tubaro [357, 358].

In Chap. 3 we employed an asymptotic expansion of the conserved energy-like quantity of a numerical method in powers of the stepsize. By positing such an expansion we are able to calculate the term of any order directly, using a recursive approach. Given such an expansion of the Hamiltonian, we have a corresponding